

# Márk Gyulai

## Senior Java Developer

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## About me

I build backend systems in Java and Kotlin. I like working on software that stays in use for years and that a team can keep developing with confidence. I also work as an AI consultant, helping companies make artificial intelligence genuinely useful.

## Experience

### Adroit Group Senior Software Engineer

October 2025 – present

As a senior software engineer I have taken part in three projects, two of them in Kotlin. I develop backends, take part in design and care about quality. One of my projects is about AI consulting.

- **MOL Limo rent a car** (Jenkins, GitLab): I develop the backend of the MOL Limo car rental system in Kotlin. The backend does not just aggregate data: it also performs calculations and synchronizes asynchronously with a subsystem.
  - Business logic and calculations in the backend, beyond aggregating data
  - Asynchronous synchronization with a subsystem
  - Managing a PostgreSQL database, including transaction management
  - Implementing a custom OAuth flow and token handling with Spring Security
  - Scheduled jobs with Spring's @Scheduled annotation
  - Documentation-first (design-first) approach
  - Writing unit and integration tests, and manual testing
  - Communicating with the client
- **Helvo** (GitLab, ArgoCD): My international team and I built a backend for frontend application for a Swiss fintech startup in Kotlin with Spring Boot 4.
  - Documentation-first (design-first) approach: designing the API in OpenAPI
  - Connecting the backend to other subsystems and the frontend with artifacts generated from the OpenAPI description
- **GLI Solutions – AI Consultant and Developer**: As an AI consultant I help artificial intelligence answer real problems. We find where it is useful, check whether it is feasible, and fit it into the systems that already exist. My team and I also work on the project as developers.
  - Developing a platform that supports manual testing, with a C# and .NET backend, a PostgreSQL database, Node.js and Angular
  - The platform can be connected to AI agents as an MCP server
  - Transaction management in the PostgreSQL database
  - Code-first approach
  - Handling bugs
  - Steering the team and the code towards clean code
  - Implementing repository-level AI skills
  - Writing documentation

Kotlin, Java, .NET, C#, Node.js, Angular, Spring Boot, Spring Security, PostgreSQL, OpenAPI, Exposed, AI, MCP, Prompting

### RDI Software Hungary Software Engineer

August 2024 – October 2025

I worked on the McDonald's application, in the team that handles user accounts, authentication and authorization. My team and I developed and maintained microservices and wrote their unit and integration tests. I also wrote SQL queries and connected other systems' APIs. Unofficially, but here too I mentored a junior colleague.

- **McDonald's** (GitHub, ArgoCD): User accounts, authentication and authorization: I was a member of the international team that handles them.
  - Maintaining existing microservices
  - Designing and developing new microservices
  - Designing API endpoints and tidying up the OpenAPI description file with reusable objects
  - Finding and fixing bugs based on metrics, traces and logs
  - Raising test coverage above 90%
  - Repairing data corrupted by migrations in a NoSQL database with a Python script
  - Managing database transactions
  - Code reviews
  - Implementing Google login with the Google SDK instead of plain API endpoints

Java, Microservices, REST API, OpenAPI, SQL, NoSQL, Python, Authentication, Authorization, Unit tests

### EPAM Systems Software Engineer

January 2022 – August 2024

As part of an international team, we developed the financial application of Broadridge, an American company. This is where I really learned multi-threaded programming, domain-driven design and the functional approach. I wrote unit, system and acceptance tests, including UI tests with Selenium. I also connected other systems' APIs, sometimes asynchronously. I solved concurrency bugs, handled database transactions and improved test coverage. I also mentored a junior developer.

- **Broadridge financial application** (Jenkins, GitLab)
- **Multitenant framework** (Jenkins, GitLab): The multitenant framework of the Broadridge project, which I worked on as a separate project.

Java, Multithreading, DDD, Multitenancy, Selenium, SQL

### EPAM Systems Junior Software Engineer

July 2021 – January 2022

As a junior developer I developed backends and learned a lot along the way. I worked on EPAM's internal exam system and later on the same Broadridge project.

- **Internal exam system** (Jenkins, GitLab): EPAM's internal exam system: a web application in which exams could be taken safely. My team and I built its backend with Spring Boot 3 and a MySQL database.
  - Designing and developing new features
  - Transaction management in the MySQL database
  - Code reviews
  - Writing unit tests with JUnit
  - Manual testing
  - Improving my skills and learning new technologies
- **Broadridge financial application** (Jenkins, GitLab): The same project I later worked on as a Software Engineer.

Java, Spring Boot 3, MySQL, JUnit, Unit tests

## Skills

AI and MCP · Java 8–25 · Kotlin · Spring Boot · Testing · Tools · Design · Angular

## Side projects

**lizacover.hu**

**This portfolio** github.com/gyulaimrk/website

## Education and certificates

**Mathematics and Computer Science, BSc**
University of Miskolc, 2017 – 2021

**Java Mentored Education Program**
EPAM Systems, September 2020 – July 2021

**English language exam, B2**

**Languages**
Hungarian, English (B2 language exam)

**Driving licence**
Category B

### Anthropic certificates

**Building with the Claude API**
Anthropic, March 2026

**Introduction to Model Context Protocol**
Anthropic, March 2026

**Introduction to agent skills**
Anthropic, March 2026

**Claude Code in Action**
Anthropic, March 2026

### More certificates (LinkedIn Learning)

**Microservices Foundations** · June 2021
**Docker for Developers** · January 2024
**Introduction to Docker for Java Developers** · January 2024
**Java 17: First Look** · January 2024
**Java: Automated API Testing with REST Assured** · January 2024
**Learning Java 17** · October 2023
**Java 8 for Professionals** · October 2023
**First Look: Java 10 and Java 11** · October 2020